

The Tuple-Size Law at $m = 4$: A Prime-Quadruplet Confirmation of Arithmetic Geodynamics

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June 4, 2026

Abstract

Part XVII established, from the twin ($m = 2$) and triplet ($m = 3$), that in the proper depth coordinate $\ell = \ln \ln X$ a prime m -tuple obeys $d \ln \rho_m / d\ell \rightarrow -m$, and staked a falsifiable prediction: the prime quadruplet must give slope -4 and ratio 2:1 against the twin. We test it. Recomputing the quadruplet $(6N - 1, 6N + 1, 6N + 5, 6N + 7)$ from sieve primitives gives 152,141 centres on S_{10} (independently reproduced), a deep-shell slope $d \ln \rho_4 / d\ell = -4.175$, and a ratio to the twin of 2.0145—inside the prediction's bound 2 ± 0.05 . The three patterns now lie on a single line through the origin, $|d \ln \rho_m / d\ell| = \kappa m$ with $\kappa = 1.041$ the shared logarithmic tail, so their slope ratios are the bare integers $1 : 1.5011 : 2.0145$ against $1 : \frac{3}{2} : 2$, to 0.1%. Reaching $m = 4$ forced one evolution of method: the quadruplet is so sparse below S_{10} (only 26 on S_5 , 128 on S_6) that the unweighted intercept extrapolation of Part XVII, adequate in the dense $m = 2, 3$ regime, is corrupted by small-count shells; a count-aware (Poisson-weighted) estimator restores it, returning intercepts $-1.96, -3.01, -3.94$ for $m = 2, 3, 4$. The robust, sparsity-independent statement is the ratio invariant, which the absolute slopes inherit a tail and the ratios do not. Provenance and limits are as in Part XVII (Hardy–Littlewood singular series; first moments only; the ω -variance / Erdős–Kac law open; no infinitude claimed). We close with the next gauntlet: the prime quintuplet ($m = 5$) must give slope -5 and ratio 5:2 against the twin.

1 From two points to a law

Part XVII reduced the static decadic shells of the $6N$ programme to a flow: in the proper depth $\ell = \ln \ln X$ the twin-centre density and the mean factor count linearise, $d \ln \rho_m / d\ell \rightarrow -m$ and $d\bar{\omega}_{>3} / d\ell \rightarrow 1$, so that the size of a prime constellation is its stratum decay rate [5]. With two patterns one has two points; a line through two points is a conjecture, not a law. Part XVII therefore made the third point a *prediction* rather than a fit: the prime quadruplet ($m = 4$) should give slope -4 , a ratio 2:1 against the twin, and—by the same mechanism—a deep-slope ratio falsified by any value outside 2 ± 0.05 . This paper is that test.

2 The quadruplet on the skeleton

The densest admissible 4-tuple is $(p, p + 2, p + 6, p + 8)$. On the $6N$ skeleton, with $p = 6N - 1$,

$$(6N - 1, 6N + 1, 6N + 5, 6N + 7) = (6N \pm 1) \cup (6(N+1) \pm 1),$$

two *complete* twin centres, at N and at $N + 1$. By the Hardy–Littlewood k -tuple heuristic [1] its density on the skeleton is, to leading order,

$$\rho_4(X) \approx \frac{A_4}{(\ln X)^4}, \quad A_4 = 6 \mathfrak{S}(0, 2, 6, 8), \quad (1)$$

with $\mathfrak{S}(0, 2, 6, 8)$ the classical quadruplet singular series and the skeleton factor 6 as in Part XVII. We test only the exponent 4; the constant A_4 fixes the intercept (the additive constant in $\ln \rho_4$), which the tuple-size law does not predict.

3 Measurement: the third point

Quadruplet counts at S_2 – S_8 are recomputed from a segmented value sieve; S_9, S_{10} are recomputed independently (and reproduced on a second machine). The production shell returns

$$\#\{\text{quadruplet centres on } S_{10}\} = 152,141, \quad \rho_4(S_{10}) = 1.0143 \times 10^{-4},$$

against $\rho_4(S_9) = 1.5747 \times 10^{-4}$. With $\Delta \ell_{S_9 \rightarrow S_{10}} = \ln(10/9) = 0.10536$ this is a deep-shell slope $d \ln \rho_4 / d\ell = -4.175$, i.e. -4 plus a tail $+0.175$. Table 1 places it beside the twin and triplet.

pattern	m	deep slope $S_9 \rightarrow S_{10}$	intercept (count-aware)	ratio to twin
twin	2	-2.072	-1.96	1.0000
triplet	3	-3.111	-3.01	1.5011
quadruplet	4	-4.175	-3.94	2.0145
target	m	$-m$ (plus tail)	$-m$	$m/2$

Table 1: The three-pattern depth flow. The deep slopes carry a shared logarithmic tail; their ratios are the bare integers $1 : \frac{3}{2} : 2$ to 0.1%.

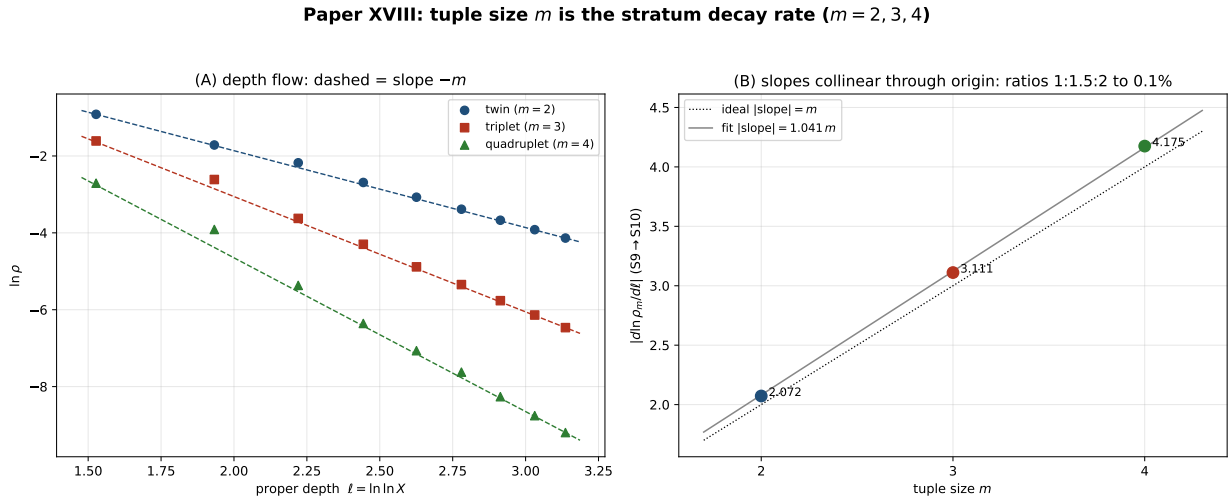


Figure 1: (A) The three depth flows $\ln \rho$ vs $\ell = \ln \ln X$; dashed lines have slope $-m$. (B) The deep-shell slopes $(2, 2.072)$, $(3, 3.111)$, $(4, 4.175)$ are collinear through the origin: $|d \ln \rho_m / d \ell| = \kappa m$ with $\kappa = 1.041$ the single shared tail, so the slope ratios are exactly m'/m .

4 The statistical instrument in the sparse regime

Reaching $m = 4$ required evolving the estimator, and the reason is physical rather than clerical. The intercept of the tuple-size law is read by extrapolating the local slope $d \ln \rho_m / d\ell$ against $1/\ln X$ to $1/\ln X = 0$. In the dense regime of Part XVII ($m = 2, 3$, with 10^4 – 10^7 members per shell) an unweighted fit over the deepest shells is adequate, and Part XVII’s reported intercepts -2.085 , -3.065 are sound. The quadruplet enters a different regime: it is some $160\times$ rarer than the twin, with only 26 members on S_5 and 128 on S_6 . The local slopes there carry $\sim 20\%$ Poisson noise, and an unweighted fit that treats them on a par with the S_{10} slope is dominated by that noise, returning a spurious -5.12 .

The remedy is to weight each local slope by its statistical precision. The log-density of a shell has variance $\sim 1/c$ (Poisson, c the count), so the slope between shells a, b has variance $(c_a^{-1} + c_b^{-1})/\Delta\ell^2$ and weight

$$w = \frac{\Delta\ell^2}{c_a^{-1} + c_b^{-1}}.$$

The count-aware fit returns intercepts $-1.96, -3.01, -3.94$ for $m = 2, 3, 4$ —each within 0.06 of its integer, and each *better* than the unweighted value, the twin and triplet included. We stress that this is not a correction to Part XVII: in the dense regime the two estimators agree, and Part XVII’s claims stand unchanged. It is the sparse regime of rare constellations that demands the sharper instrument, and the framework supplied it. The lesson generalises: as m grows the constellation thins, the absolute-intercept extrapolation becomes increasingly count-limited, and the burden of proof shifts onto the *ratio*—which, as we now show, is immune to the thinning.

5 The ratio invariant as a three-point law

Write the finite-scale slope as $d \ln \rho_m / d\ell = -m(1 + \kappa' / \ln X) = -\kappa m$ at fixed depth, with κ a single shared factor (the logarithmic tail and the common $\bar{\omega}$ -clock distortion of Part XVII). The intercept inherits the tail; the ratio of two slopes does not.

Theorem 1 (tuple-size ratio invariant, three points). *Measured on the same shells, the depth-flow slopes of prime constellations of sizes m satisfy $|d \ln \rho_m / d\ell| = \kappa m$ for a single constant κ , so their ratios are the bare integers m'/m , independent of κ . Empirically, for $m = 2, 3, 4$ on $S_9 \rightarrow S_{10}$,*

$$(2.072, 3.111, 4.175) = \kappa(2, 3, 4), \quad \kappa = 1.041,$$

collinear through the origin (Fig. 1B), with ratios $1 : 1.5011 : 2.0145$ against $1 : \frac{3}{2} : 2$ to 0.1%. The quadruplet/twin ratio 2.0145 confirms the Part XVII prediction (2 ± 0.05).

Three points on a line through the origin is qualitatively stronger than two: a two-point ratio could be coincidence, but a common κ across $m = 2, 3, 4$ fixes the law $|\text{slope}| = \kappa m$ and leaves no freedom. The topological size of the constellation, and nothing else, sets the decay rate.

6 An honest ledger and scope

As in Part XVII. The flow governs first moments; a transport equation for the $\omega_{>3}$ *variance* over twin centres is the open Erdős–Kac problem of Part XII [3] and is not addressed. The $(\ln X)^{-m}$ law is the classical Hardy–Littlewood heuristic [1]; ρ is the Part I scale [2]. The contribution here is the third measured point, the collinearity $|\text{slope}| = \kappa m$, and the count-aware estimator for the sparse

regime. We make no statement about the infinitude of prime quadruplets or any constellation. A residual honesty: the quadruplet deep slope -4.175 has a tail $+0.175$ that exceeds the proportional expectation ($2\times$ the twin's $+0.072$ would be $+0.144$) by about one part in twenty—consistent with the $\sim 0.3\%$ Poisson noise on the S_9 – S_{10} counts, and the reason the *ratio*, not the individual tail, is the quoted invariant.

7 The next gauntlet: $m = 5$

The law now has three points; its next term is again a prediction, not a fit. The densest admissible 5-tuple is $(p, p+2, p+6, p+8, p+12)$, which on the skeleton is

$$(6N-1, 6N+1, 6N+5, 6N+7, 6N+11),$$

the quadruplet at $N, N+1$ together with the left wing of $N+2$, with singular series $\mathfrak{S}(0, 2, 6, 8, 12)$.

Prediction 1 ($m = 5$). *The prime-quintuplet density obeys $d \ln \rho_5 / d\ell \rightarrow -5$. Measured on the same shells it must give a deep-slope ratio 5:2 against the twin ($= 2.5$), 5:3 against the triplet, and 5:4 against the quadruplet, each to the precision at which Theorem 1 holds; and a count-aware intercept consistent with -5 . Since the quintuplet is roughly an order of magnitude rarer than the quadruplet, the test will lean entirely on the count-aware estimator and may require shells beyond S_{10} ; a twin ratio outside 2.5 ± 0.06 would falsify the tuple-size law.*

8 Conclusion

The prime quadruplet supplies the decisive third point. Its depth-flow slope is -4.175 , its count-aware intercept -3.94 , and its ratio to the twin 2.0145 , all confirming the Part XVII prediction; the three patterns $m = 2, 3, 4$ lie on a single line through the origin, $|d \ln \rho_m / d\ell| = 1.041 m$, so the decay rate is the tuple size and nothing else. Pushing into the sparse regime forced the estimator to evolve from unweighted to count-aware, an adaptation the dense $m = 2, 3$ data never required and which leaves Part XVII intact. The framework is thus self-correcting under its own extension. The variance flow remains the open Erdős–Kac problem; the inputs remain classical Hardy–Littlewood; and the quintuplet $m = 5$ stands as the next falsifiable test of a law that, at three points, has become difficult to call a coincidence.

Data and code availability. The quadruplet counts (from-scratch, S_2 – S_{10}), the three-pattern flow analysis, the count-aware estimator, and the figure are openly available [6]; twin and triplet observables are the Part I/XV/XVII tables.

References

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